# Chapter Two

## Problem Solving

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# Syllabus

2.1.Defining problems as a state space search,

2.2.Problem formulation

2.3.Problem types, Well- defined problems, Constraint satisfaction problem,

2.4.Game playing, Production systems

**Credit Hours: 4 hrs**

**Marks: 7**

# What is problem?

**Noun**: a **matter or situation** regarded as **unwelcome or harmful** and needing to be **dealt with and overcome.**

**Adjective**: **denoting or relating to people** whose behaviour causes **difficulties to themselves and others**

# Problem Solving in AI

- ★ Problem solving in Artificial Intelligence (AI) refers to the process by which an AI system **identifies, formulates, and solves problems.**
- ★ AI problem-solving techniques typically involve **defining a problem in terms of states and actions**, and then **searching through a space of possible solutions** (called the **state space**) to find the **best solution.**
- ★ This **approach** is foundational in artificial intelligence (AI) and **helps structure how we think** about and **solve complex problems.**
- ★ Problem solving is a **agent based system** that finds **sequence of actions** that lead to **desirable states from the initial state.**

# Problem Solving in AI

- ★ Problem solving, particularly in artificial intelligence, may be characterized as a **systematic search through a range of possible actions** in order to reach some **predefined goal or solution.**
- ★ Problem-solving methods divide into **special purpose and general purpose.**
- ★ A **special purpose method** is tailor-made **for a particular problem** and often exploits very **specific features of the situation** in which the problem is embedded.
- ★ In contrast, a **general-purpose** method is applicable to a **wide variety of problems.**

# Problem formulation

A problem is defined by:

★ **An initial state**: State from which **agent start**
★ **Successor function**: Description of **possible actions available** to the agent.
★ **Goal test:** Determine whether the given state is **goal state or not**
★ **Path cost:** Sum of **cost of each path** from initial state to the given state.

➔ A **solution** is a **sequence of actions** from **initial to goal state.**
➔ **Optimal** solution has the **lowest path cost.**

# State Space Search

- ★ To **locate a solution**, state space search entails **methodically** going through **every potential state** for an issue.
- ★ This approach can be used to solve a variety of AI issues, including **pathfinding, solving puzzles, playing games**, and more

# State Space Representation

- **States**: Different arrangements of the issue, frequently shown as graph nodes.
- **Initial State**: The initial setting that the search starts with.
- **Goal State(s)**: The ideal configuration(s) denoting a resolution.
- **Actions**: The processes by which a system changes states.
- **Transition Model**: Explains what happens when states are subjected to actions.
- **Path cost:** The expense of moving from an initial state to a certain state, expressed as a numerical value linked to each path.

# Example - 8 Puzzle Problem

**States**: Every arrangement of the 3x3 grid consisting of tiles numbered 1 through 8 and a blank area.

**Initial State**: A particular tile layout at the outset.

**Goal State**: The configuration with the blank space in the lower-right corner and the tiles arranged in numerical order.

**Actions**: Up, down, left, or right movement of the empty area.

**Transition Model:** Describes the state that arises after carrying out a certain action.

**Path Cost:** The uniform cost of every motion is one.